

Knowing less, exploring more? Metacognitive regulation of knowledge in explore-exploit decisions

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Abstract

Explore-exploit decisions are commonly guided by individuals' knowledge states, with accumulating knowledge typically reducing exploration and promoting a shift toward exploitation. However, beyond the amount of knowledge one possesses, metacognitive regulation of knowledge may critically shape exploration. This study investigates how metacognitive regulation of one's knowledge base influences explore-exploit decisions and interacts with individuals' actual knowledge levels. Experiment 1 showed that, contrary to the expert–novice hypothesis that limited knowledge promotes exploration, individuals with greater prior knowledge exhibited higher exploration than those with less prior knowledge. This effect was accompanied by greater awareness of one's own knowledge state among more knowledgeable individuals. Experiment 2 provided converging evidence by showing that metacognitive regulation through self-monitoring further increased exploration. Together, these findings suggest that explore–exploit decisions are shaped not only by how much individuals know, but also by how they metacognitively regulate their knowledge.

Keywords: explore-exploit decisions; prior knowledge; metacognitive regulation; self-monitor

Introduction

Decisions between exploration and exploitation apply widely across diverse domains of daily life, requiring individuals to choose between exploring unknown options versus exploiting familiar ones (Mehlhorn et al., 2015; Cohen et al., 2007; Wilson et al., 2021; Hills et al., 2015; Harms et al., 2024). For example, when the ice cream shop that offers our favorite flavor introduces a new one, we find ourselves tense between choosing the familiar favorite and trying the new flavor. On the one hand, more exploitative options tend to lead to a steady reward level but may result in the agent being stuck in a “good enough” state due to the more constrained nature of such options. On the other hand, a broader and more exploratory search strategy may increase the potential for achieving a better outcome but can also create higher uncertainty at the same time (Wu et al., 2018; Wilson et al., 2014; Schulz and Gershman, 2019).

In explore–exploit decision-making contexts aimed at reward maximization, individuals typically begin with exploration and gradually shift toward exploitation as knowledge about the environment accumulates (Wilson et al., 2014; Blanco and Sloutsky, 2021). The early exploratory phase serves as a period of learning, during which individuals form expectations about potential outcomes (Wu et al., 2018;

Schulz et al., 2019). The reduced uncertainty ultimately enables individuals to drive the explore-to-exploit shift (e.g., adults' behaviors in Blanco and Sloutsky, 2021; Liquin and Gopnik, 2022). Research comparing experts and novices consistently demonstrates that greater prior knowledge corresponds to reduced exploration and more efficient solutions across diverse domains (e.g., physicians, Reyna and Lloyd, 2006; police, Garcia-Retamero and Dhami, 2009; teachers, Carter et al., 1988). Experts achieve higher performance by selectively attending to fewer, more diagnostic information dimensions, thereby minimizing redundant or diffuse exploration (Gigerenzer and Gaissmaier, 2011; Anders Ericsson and Towne, 2010). Together, these findings suggest that accumulated knowledge shortens the exploration phase, promotes exploration efficiency, and accelerates the shift from exploration to exploitation.

Although greater knowledge tends to reduce exploration due to increasing certainty about situations, larger knowledge gaps do not necessarily elicit higher levels of exploration. Wade and Kidd (2019) showed that learners' motivation to explore is strongest when they believe they are close to the correct answer. Connecting to the “region of proximal learning (RPL)” proposed by Metcalfe et al. (2020), the best mental state to trigger exploration needs an adaptive mental space where we feel we are on the verge of knowing. These findings indicate that exploration is not directly determined by factual knowledge levels or proportional to knowledge deficits. Instead, exploration is more likely to arise when individuals believe that engaging in exploration will yield effective knowledge gains (Kidd et al., 2012; Kang et al., 2009; Loewenstein, 1994). These findings converge on the idea that exploration is guided by learners' awareness of their current knowledge position, which involves the metacognitive regulation of learning processes (Goupil and Proust, 2023).

Metacognitive regulation serves as a key learning tool that actively drives knowledge acquisition and self-regulated learning (e.g., Akturk and Sahin, 2011; Zepeda et al., 2015; Veenman et al., 2006; Schraw, 1998; Flavell, 1979). It comprises key processes—planning, monitoring, and evaluation—that enable learners to select effective strategies, allocate attention, and flexibly adjust behavior in response to task demands (Schraw, 1998). Evidence shows that in the task of answering an uncertain question after reading an article,

learners with metacognitive awareness reflected more strategically and performed better than unaware learners (Garner and Alexander, 1982; Joseph et al., 2016). Extending the correlation findings, experimental work demonstrates that explicit training in metacognitive regulation enhances learners’ monitoring skills and transfers to novel self-regulated learning tasks (Schuster et al., 2020). Together, these findings highlight metacognitive regulation as a central mechanism through which learners actively guide their decisions and behaviors. However, little is known about how actual knowledge bases interact with metacognitive regulation of one’s knowledge state to shape exploration preferences.

Overall, the present study investigates how metacognitive regulation of one’s knowledge base shapes explore–exploit decisions and how it interacts with the amount of actual knowledge possessed. To address this question, Experiment 1 manipulated participants’ prior factual knowledge about the task, creating conditions with relatively less versus more initial knowledge base, in order to isolate the effect of knowledge base per se on explore-exploit decisions. Building on this design, Experiment 2 introduced an explicit metacognitive manipulation that encouraged participants to reflect on their own knowledge state through an ongoing self-monitoring of knowledge state during the task. By jointly considering these two Experiments, it enables us to dissociate the effects of metacognitive regulation of knowledge from the effects of knowledge amount alone. We hypothesized that promoting self-monitoring would increase attention to one’s knowledge state—such as perceived knowledge gap and progress—and that heightened awareness of knowledge deficits would, in turn, encourage higher exploration.

Experiment 1

Experiment 1 was to examine whether different explore-exploit decisions are evident among individuals with varied actual knowledge base. Here, participants were exposed to either minimal (actual less knowledge) or relatively complete information (actual more knowledge) about the task. Participants were not explicitly informed about their relative knowledge base.

Participants

85 students from several universities were recruited online, aged from 18 to 30 year olds. All participants signed assent forms before the experiment.

Design

Participants were randomly divided into two groups: the **Actual-Less** group or the **Actual-More** group. The Actual-Less group ($n = 42$, 47.62% female; age range from 19.3 to 30.2, mean age = 21.7) received no additional contextual information other than the basic rules and goal of the game. The Actual-More group ($n = 43$, 48.84% female; age range from 18.6 to 25.7, mean age = 21.9) was additionally provided with details about the characteristics of the search environment in the task.

Task

The explore-exploit task contains a clicking game for participants to search and collect rewards, which was revised from a spatially correlated multi-armed bandit paradigm originally presented by Wu et al. (2017). In our experiment, we present participants with a total of 64 tiles in an 8×8 grid world (see Figure 1). Each tile contains a certain number of rewards, with the reward ranging from 1 to 32. The reward of a tile is assigned following a “smooth” rule, which states that nearby tiles have similar rewards. The tiles have different colors depending on the reward they contain, with tiles containing greater rewards having darker colors. The rewards of tiles were hidden from participants initially, with all the tiles displaying a gray color. Participants can interact with this grid world by clicking on one tile at a time, revealing and collecting the reward of that tile. When a tile is clicked on for the first time, its color changes from gray to its assigned color, which signifies its reward. The tiles that have already been clicked on will remain in their assigned color and can be clicked on again. After every click, participants receive information about the amount of reward they received in this click, the total amount of reward they have received, and the number of clicking opportunities left for this round. Participants were told that they needed to engage in 3 rounds of the game, aiming to maximize their total rewards within 25 clicks per round. The distribution of rewards is different in these 3 rounds. In this task, participants can click on a new tile (exploration) or re-click an already-revealed one (exploitation), representing two distinct options in the explore-exploit decision process. Participants’ behavioral patterns, as well as the total rewards they earned, are assessed.

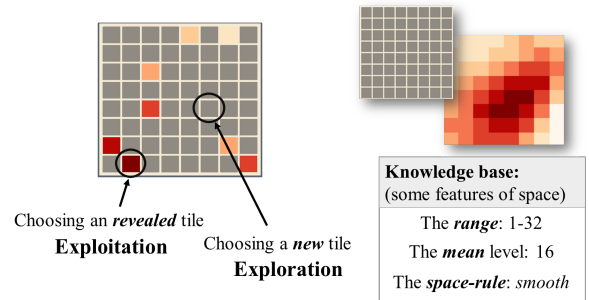


Figure 1: Explore-exploit task. Participants search and collect rewards in an 8×8 grid world. Darker-colored tiles have greater rewards. Participants could choose a new tile (exploration) or choose a revealed one (exploitation).

Procedure

Instruction phase. All participants began with standardized instructions explaining that they would search for and collect rewards hidden behind 64 tiles arranged in an 8×8 grid. Two core rules were emphasized: (1) darker tile colors indicated higher rewards, and (2) each tile could be re-clicked

multiple times. Participants were then informed of the task structure and goal: (1) there are 3 rounds of searching games, and each round provides 25 clicking opportunities. (2) Before completing the 3 experimental rounds, there is a free round of practice trials. (3) The task goal is to accumulate as many total rewards as possible.

Following this general introduction of basic rule and goal of the task, the procedure diverged by experimental groups. Participants in the Actual-Less group received no additional information about the structure of the search environment. In contrast, participants in the Actual-More group were provided with three additional pieces of information about the characteristics of the search environment: the reward range, the reward mean, and the spatial rule of reward distribution. Specifically, they were informed: “The values of rewards range from 1 to 32, and the mean value is 16. The existing values of rewards exhibit a smooth, gentle fluctuation. And the distribution of these rewards in such environments is regular and smooth, so that closer tiles would likely have more similar rewards.”

Play phase. Participants first completed a free practice round, followed by three task rounds in which they searched for and collected rewards. Each round allowed up to 25 clicks, with no time constraints imposed.

Post-task phase. Upon completion of the explore-exploit task, participants completed a post-task questionnaire designed to assess their subjective experience during the task, including self-assessed knowledge state, confidence in task performance, and attention to the knowledge state and performance.

Results

Exploration ratio. To assess participants’ exploration preference across two groups, we first examined their exploration ratio in each round. The exploration ratio was defined as the percentage of chosen tiles that were new (number of exploration choices / 25). We conducted a linear mixed-effects model with random subject effects, including fixed effects of knowledge base, round, and their interaction, alongside age and gender as covariates. Results indicate that participants with actual less knowledge ($M = 45.90\%$, $SD = 24.67\%$) had a significantly lower exploration ratio than those who with actual more knowledge ($M = 54.98\%$, $SD = 30.47\%$), $\beta = 0.12$, 95% CI [2.62e-03, 0.24], $t(245) = 2.01$, $p = 0.045$ (see Figure 2A). This finding indicates that individuals with a greater prior knowledge, rather than those who had a smaller knowledge base, tend to engage more in exploration. No effect of round across these two groups is found (see Figure 2B). This may suggest that there was no strong familiarity effect in this task, as increased exposure to the search environment did not lead to reduced exploration.

Search choices. To understand individuals’ exploration patterns of the two groups in more detail, we calculated the distance between each pair of consecutive choices and classi-

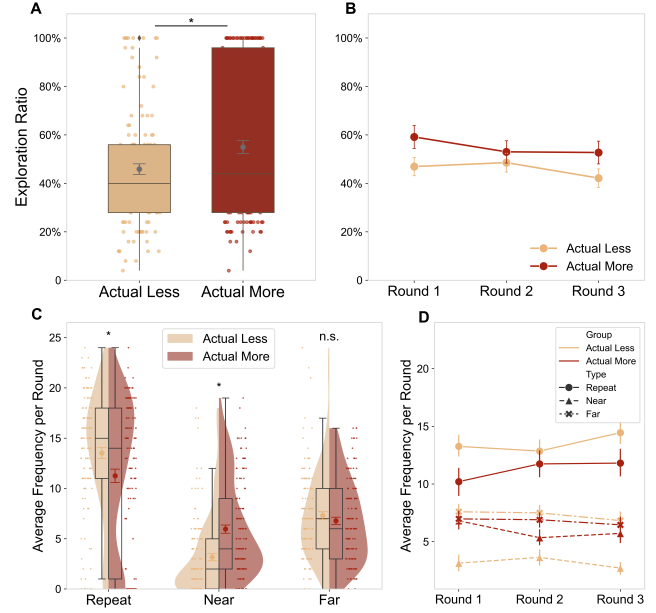


Figure 2: (A) The average exploration ratio in Actual-Less and Actual-More groups. Individuals with less prior knowledge explored less than those with more prior knowledge. (B) Exploration ratio across 3 rounds for both groups. (C) The average frequency of 3 types of search choice types (Repeat, Near, and Far) in both groups. Compared to individuals with less prior knowledge, individuals with more knowledge base made fewer Repeat choices and more Near choices. (D) Frequencies of the 3 choice types across rounds for both groups.

fied each choice into one of three types: repeat, near, and far. Repeat choices: the individual clicked the exact same tile as in the previous click (distance = 0). Near choices: the individual clicked a tile in the immediately adjacent neighborhood of the previous click (distance = 1 and $\sqrt{2}$). Far choices: the individual clicked a more distant tile (distance $> \sqrt{2}$). We then tested the differences in group frequency of these 3 types of choices by the linear mixed-effects models. The results reveal that participants with less prior knowledge made significantly more repeat choices ($M = 13.52$ times, $SD = 6.17$) than those with more prior knowledge ($M = 11.26$ times, $SD = 7.62$), $\beta = -3.05$, 95% CI [-6.04, -0.07], $t(245) = -2.01$, $p = 0.045$, mirroring the pattern observed for exploration ratio. In contrast, participants with less prior knowledge made fewer near choices ($M = 3.17$ times, $SD = 3.97$) than participants with more prior knowledge ($M = 5.96$ times, $SD = 4.68$), $\beta = 3.69$, 95% CI [1.87, 5.51], $t(245) = 3.99$, $p < 0.001$, with this effect also stable across rounds. Finally, participants in both groups made a very similar number of Far choices (actual-less: $M = 7.31$ times, actual-more: $M = 6.78$ times), $\beta = -0.63$, 95% CI [-2.47, 1.21], $t(245) = -0.68$, $p = 0.500$ (see Figure 2C and D). Overall, compared to individuals who have actual less knowledge, individuals with more knowledge made fewer repeat choices and more near choices.

Reward performance. No significant difference in reward performance per round was observed between participants who had a less knowledge base ($M = 634.39$, $SD = 86.74$) and those who had a more knowledge base ($M = 638.64$, $SD = 93.50$), $\beta = -7.12$, 95%CI [-46.20, 31.97], $t(245) = -0.36$, $p = 0.720$. Participants’ reward performance was comparable across rounds for both groups.

Post-task responses. Participants in the Actual-More group reported slightly higher self-assessed knowledge about the search environment ($M = 3.42$, $SD = 0.82$) than those in the Actual-Less group ($M = 3.14$, $SD = 0.57$), although this difference did not reach conventional levels of statistical significance, $t(74.60) = -1.80$, $p = 0.075$, $d = -0.42$. Interestingly, participants in the Actual-More group reported somewhat greater attention to their knowledge of the search environment ($M = 3.19$, $SD = 1.62$) than those in the Actual-Less group ($M = 2.52$, $SD = 1.89$). Specifically, attention to one’s knowledge state reflects an awareness of what is known versus unknown, enabling individuals to monitor and regulate the dynamics of their knowledge over time. This awareness may function as an initial “gatekeeper”, through which perceived knowledge sufficiency or insufficiency shapes subsequent exploration preferences. In this way, metacognitive regulation of the knowledge base plays a central role in guiding explore–exploit decisions. This perspective directly motivates the design of Experiment 2, which aims to more explicitly examine this underlying mechanism.

Experiment 2

In Experiment 2, we examined how metacognitive regulation of knowledge influences explore–exploit decision-making. To this end, we introduced a manipulation that prompted participants to actively self-monitor their knowledge throughout the task, thereby increasing metacognitive awareness of their momentary knowledge state.

Participants

76 students from several universities were recruited online, aged from 18 to 30 year olds. All participants signed assent forms before the experiment.

Design

Participants were randomly assigned to one of two conditions: the **Monitor-Actual-Less** group or the **Monitor-Actual-More** group. The Monitor-Actual-Less group ($n = 45$, 51.11% female; age range from 18.73 to 25.75, mean age = 21.31) received no additional information about the characteristics of search environment, mirroring the Actual-Less group in Experiment 1. Additionally, participants in this group engaged in an explicit metacognitive regulation by self-monitoring of their current knowledge state and perceived learning progress at the end of each round. The Monitor-Actual-More group ($n = 31$, 51.61% female; age range from 18.86 to 25.25, mean age = 20.86) received detailed information about the task structure and reward pat-

terns, analogous to the Actual-More group in Experiment 1. Additionally, unlike participants in Experiment 1, those in both groups of Experiment 2 engaged in explicit metacognitive regulation by actively monitoring their current knowledge state and perceived learning progress at the end of each round.

Task

The task was identical to that used in Experiment 1.

Metacognitive regulation manipulation

Participants in both groups were required to engage in self-monitoring of their knowledge by assessing their current knowledge (Perceived-Knowledge-State) and predicting their potential progress (Expected-Knowledge-Progress) using a 5-point Likert scale (see Table 1 for specific items). Both groups completed identical versions of this scale. Because the task included a practice round, the first self-reported scores collected after this practice round were treated as participants’ baseline perceived knowledge state and expected knowledge progress prior to the first experimental round. In the analysis, we examined the association between such previously self-monitoring of knowledge and subsequent exploration preferences.

Table 1: Metacognitive regulation manipulation by self-monitoring of knowledge (1-5 point scale)

(Perceived-Knowledge-State) Based on current experience, how well do you think you know the features of the search environment now?
(Expected-Knowledge-Progress) How much will you be able to improve your understanding of the features of the search environment in the next round?

Procedure

Participants in the Monitor-Actual-Less and Monitor-Actual-More knowledge conditions followed the same overall procedure. Unlike Experiment 1, participants in this Experiment were required to self-assess and monitor their knowledge state and progress during the play phase. Figure 3 illustrates the specific procedure.

Instruction phase. The instructions for the Monitor-Actual-Less (More) group mirrored those of the Actual-Less (More) group in Experiment 1 with respect to the task’s rules and goals. Participants were additionally informed that they would report several items after each round. To minimize demand characteristics, the purpose of these reports—namely, monitoring their knowledge state—was not made explicit.

Play phase with self-monitoring of knowledge. Participants first completed a practice round, followed by three experimental rounds, during which they assessed their knowl-

edge state and predicted their learning progress after each round.

Post-task phase. This phase was identical in Experiment 1.

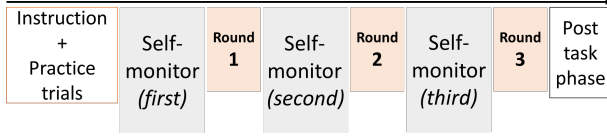


Figure 3: Procedure in Experiment 2. Participants were required to monitor their knowledge state during the task.

Results

Exploration ratio. We conducted identical analysis like in Experiment 1 to compare participants' exploration ratio across two groups with metacognitive regulation of their knowledge. There was no main effect of knowledge base ($\beta = -0.04$, $p = 0.525$), indicating that overall exploration levels did not differ between groups. In contrast, there was a main effect of round, such that exploration decreased over time: compared to Round 1, exploration ratio was lower in Round 2 ($\beta = -0.06$, $p = 0.045$) and further reduced in Round 3 ($\beta = -0.10$, $p < 0.001$).

Search choices. For *repeat choices*, results mirrored with the ones in exploration ratio. There was no main effect of knowledge base ($\beta = 1.09$, $p = 0.525$), whereas there was a main effect of round: compared to Round 1, repeat choices increased in Round 2, $\beta = 1.53$, $p = 0.045$; and further increased in Round 3, $\beta = 2.60$, $p < 0.001$ (Monitor-Actual-Less: $M_s = 9.67, 11.20, 12.27$ by rounds; Monitor-Actual-More: $M_s = 10.61, 10.90, 11.00$ by rounds). For *near choices*, results indicated relatively stable patterns across rounds in both groups (Monitor-Actual-Less: $M_s = 5.22, 4.84, 4.67$; Monitor-Actual-More: $M_s = 6.06, 6.00, 5.90$). Neither the main effects of knowledge base ($\beta = 0.97$, $p = 0.333$) and round (Round 2: $\beta = -0.38$, $p = 0.502$; Round 3: $\beta = -0.56$, $p = 0.324$), nor their interaction (all $p > 0.65$) were significant. For *far choices*, there was no main effect of knowledge base ($\beta = -2.06$, $p = 0.084$), although a marginal trend suggested fewer far choices in the Monitor-Actual-More group. Such far choices decreased over time (Monitor-Actual-Less: $M_s = 9.11, 7.96, 7.07$ by rounds; Monitor-Actual-More: $M_s = 7.32, 7.10, 7.10$ by rounds), reaching significance in Round 3 ($\beta = -2.04$, $p = 0.002$). Together, results reveal a consistent search dynamics over time across two groups: participants increasingly relied on repeat choices while reducing far choices, with near choices remaining relatively stable. Crucially, these dynamics suggested that under metacognitive regulation, changes in search strategy primarily reflect within-task adaptation rather than between-group differences in knowledge.

Reward performance. There was no significant main effect of knowledge base ($\beta = 37.66$, $p = 0.084$), although the

Monitor-Actual-More group showed a marginal tendency toward higher rewards. In contrast, reward performance increased reliably across rounds: compared to Round 1, rewards were higher in Round 2 ($\beta = 34.47$, $p = 0.007$) and further increased in Round 3 ($\beta = 47.53$, $p < 0.001$). Overall, these findings suggest that reward gains primarily reflected within-task learning and adaptation rather than differences in initial knowledge base under metacognitive regulation.

Dynamics in self-monitoring of knowledge. Under metacognitive regulation, participants with less prior knowledge reported greater *expected knowledge progress* ($M = 3.39$, $SD = 1.00$) than those with greater prior knowledge ($M = 3.03$, $SD = 1.02$), $\beta = -0.67$, $p = 0.003$. Interestingly, participants with less prior knowledge also reported higher *perceived knowledge state* about the search environment ($M = 4.18$, $SD = 1.00$) than those with greater prior knowledge ($M = 3.96$, $SD = 1.02$), $\beta = -0.52$, $p = 0.008$. Together, these findings underscore the role of metacognitive processes—such as perceived knowledge state and anticipated knowledge progress—in shaping explore-exploit decisions.

Furthermore, we fitted a linear mixed-effects model predicting exploration ratio from perceived knowledge state, expected knowledge progress, and round, controlling for age and gender, with random intercepts for subjects. Within this model, expected knowledge progress was a positive predictor of exploration ratio ($\beta = 0.03$, $p = 0.023$), indicating that participants who expected greater knowledge progress tended to engage in more exploration, which is consistent with the learning progress theory (Oudeyer et al., 2016). In contrast, perceived knowledge state did not significantly predict exploration ($\beta = -0.003$, $p = 0.862$).

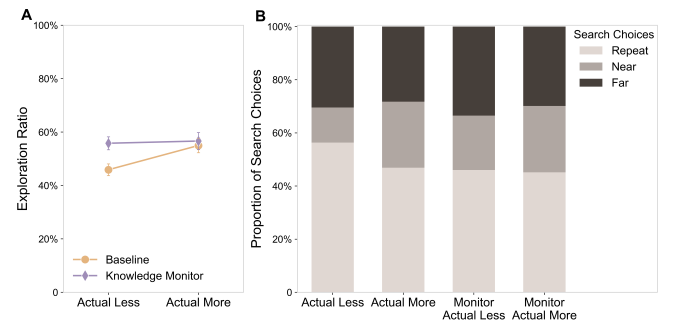


Figure 4: (A) Exploration ratio across less and more actual knowledge base in the Baseline conditions (Exp 1) and the Knowledge-Monitor conditions (Exp 2). (B) Proportions of 3 search choice types across less and more actual knowledge base in the Baseline conditions (Exp 1) and the Knowledge-Monitor conditions (Exp 2).

Experiment 1 (Baseline) & 2 (Knowledge Monitor)

We examined how actual knowledge base and metacognitive regulation (self-monitoring of knowledge) jointly shaped explore-exploit decisions across rounds.

Results showed that participants with greater knowledge exhibited higher exploration than those with less knowledge ($\beta = 0.12$, $p = 0.048$), and participants in the knowledge-monitor condition also showed greater exploration than those in the baseline condition ($\beta = 0.14$, $p = 0.024$). Thus, both actual knowledge and self-monitoring independently promoted exploration. The interaction between the two factors was marginally significant, $\beta = -0.16$, $p = 0.074$ (see Figure 4A), suggesting that metacognitive regulation may attenuate knowledge-based differences in exploration preference.

For proposition of 3 search choices (repeat, near, and far), multinomial logistic regression showed that participants with greater knowledge base were more likely to make near choices rather than repeat choices, $\beta = 0.82$, $z = 11.45$, $p < 0.001$, and metacognitive regulation further increased this tendency, $\beta = 1.24$, $z = 7.71$, $p < 0.001$ (see Figure 4B). However, the interaction was negative, $\beta = -0.60$, $z = -5.97$, $p < 0.001$, indicating that metacognitive regulation attenuated knowledge-based differences in local exploration. For far choices (relative to repeat), knowledge base showed only a marginal effect, $\beta = 0.11$, $z = 1.84$, $p = 0.066$, whereas metacognitive regulation significantly increased far exploration, $\beta = 0.50$, $z = 3.82$, $p < 0.001$. Again, a negative interaction emerged, $\beta = -0.20$, $z = -2.35$, $p = 0.019$, suggesting reduced choice differences between knowledge groups under metacognitive regulation. Overall, both greater knowledge base and metacognitive regulation promoted exploration over repetition, but metacognitive regulation compressed behavioral differences across knowledge base. Near exploration appeared to reflect efficient, knowledge-guided local search, whereas distant exploration was less knowledge-dependent and more strongly associated with broader exploration strategies under metacognitive regulation.

General Discussion

How does metacognitive regulation of one's knowledge base influence explore-exploit decisions, and how this process interacts with individuals' actual levels of prior knowledge? In Experiment 1, individuals with greater actual prior knowledge showed higher levels of exploration than those with less prior knowledge. This pattern contrasts with the expert-novice hypothesis, which predicts that limited knowledge should promote exploration. Thus, our results in Experiment 1 highlighted that exploration is not determined solely by the amount of knowledge one possesses (Loewenstein, 1994; Oudeyer et al., 2016). Importantly, post-task reports revealed that individuals with greater prior knowledge also attended more to their own knowledge state during the task, pointing to a potential role of metacognitive regulation in shaping exploration preferences.

Experiment 2 directly manipulated metacognitive regulation by requiring participants to monitor their knowledge—self-assess their knowledge state and predict their knowledge progress during the task. Building on this insight, this manipulation selectively increased exploration among individu-

als with less prior knowledge, relative to their counterparts who did not engage in self-monitoring (Experiment 1). Individuals with greater prior knowledge maintained high levels of exploration regardless of whether self-monitoring was introduced. One plausible explanation is that these individuals were already more attuned to their knowledge state, as the post-task attention-to-knowledge measures observed in Experiment 1. Taken together, these findings support the conclusion that metacognitive regulation of one's knowledge—specifically, awareness of what is known and unknown—serves as a key internal mechanism driving exploration and shaping explore-exploit decisions (Ten et al., 2021).

From a goal-orientation perspective on behavioral motivation (Dweck, 1986; Molinaro and Collins, 2023), metacognitive regulation induced through self-monitoring of knowledge may implicitly activate an information-seeking goal, thereby increasing motivation to explore. In typical explore-exploit contexts, the dominant and salient goal is reward maximization, whereas information seeking is less explicit as a guiding objective. By heightening awareness of one's knowledge state, metacognitive regulation likely makes learning and information acquisition more salient, reinforcing exploration.

Interestingly, the dynamics in self-monitoring of knowledge in Experiment 2 revealed systematically cognitive biases in one's self-assessment, with participants' perceived knowledge states deviating from their actual knowledge. In particular, individuals with less actual knowledge tended to exhibit overconfidence. Prior research suggests that self-monitoring and self-evaluation are highly susceptible to cognitive biases (e.g., Nickerson, 1998; Sharot, 2011), frequently leading individuals to overestimate their performance and abilities (Miller and Geraci, 2011; De Bruin et al., 2017). This pattern is classically captured by the Dunning-Kruger effect, whereby less competent individuals substantially overestimate their abilities, whereas more competent individuals tend to underestimate their own competence (Kruger and Dunning, 1999; Dunning, 2011).

Overall, this study demonstrates that explore-exploit decisions are shaped not simply by how much individuals truly know, but by how they metacognitively regulate knowledge. By disentangling actual prior knowledge from metacognitive regulation, we show that awareness of one's knowledge state is a critical mechanism that promotes exploration, particularly when prior knowledge is limited. These findings challenge knowledge-deficit accounts of exploration and underscore the importance of metacognitive regulation in adaptive explore-exploit decisions. This work suggests that interventions targeting metacognitive awareness—rather than knowledge accumulation alone—may more effectively foster adaptive exploration behaviors, with important implications for everyday decision-making, learning, and education. Future research can extend this framework by examining how different forms of metacognitive regulation, goal framing (e.g., learning-oriented vs. reward-oriented), and developmental factors interact to shape explore-exploit decisions.

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